FILTER DESIGN TEST CASE

Band-Pass Filter - WiFi 2.4 GHz for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting WiFi 2.4 GHz for Satellite L-Band Filter. The filter is designed on 42° YX LiTaO3 substrate with a target center frequency of 3169.5 MHz and bandwidth of 207.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: WiFi 2.4 GHz
- Application: Satellite L-Band Filter
- Substrate: 42° YX LiTaO3
- Package: SiP (System in Package)
- Design Tool: PathWave EM Design
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Band-Pass Filter design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 3169.5 MHz
Bandwidth (3 dB): 207.6 MHz
Insertion Loss Target: <= 3.6 dB
Return Loss Target: >= 14.0 dB
Out-of-Band Rejection: >= 47.0 dB
Isolation (Duplexer): >= 42.0 dB
Q Factor: 3221
Temperature Coefficient:-10.2 ppm/°C
Die Size: 1.1 x 0.8 mm
Wafer Size: 6-inch
Number of Resonators: 11
Electrode Material: Mo/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Band-Pass Filter template project.
2. Define the substrate stack: 42° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 3169.5 MHz, BW 207.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.6 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 42° YX LiTaO3).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9508 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.6 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 311 MHz offset >= 47.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 4.0 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-07-23
Devices Tested: 54
Insertion Loss (meas): 3.35 dB
Return Loss (meas): 16.1 dB
Rejection (meas): 50.0 dB
Isolation (meas): 46.5 dB
Q Factor (meas): 3221
Group Delay Variation: 1.2 ns
Power Handling: +29 dBm
Die Yield: 91.7%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The Band-Pass Filter design demonstrated below-target performance for WiFi 2.4 GHz.
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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